I. WAVES

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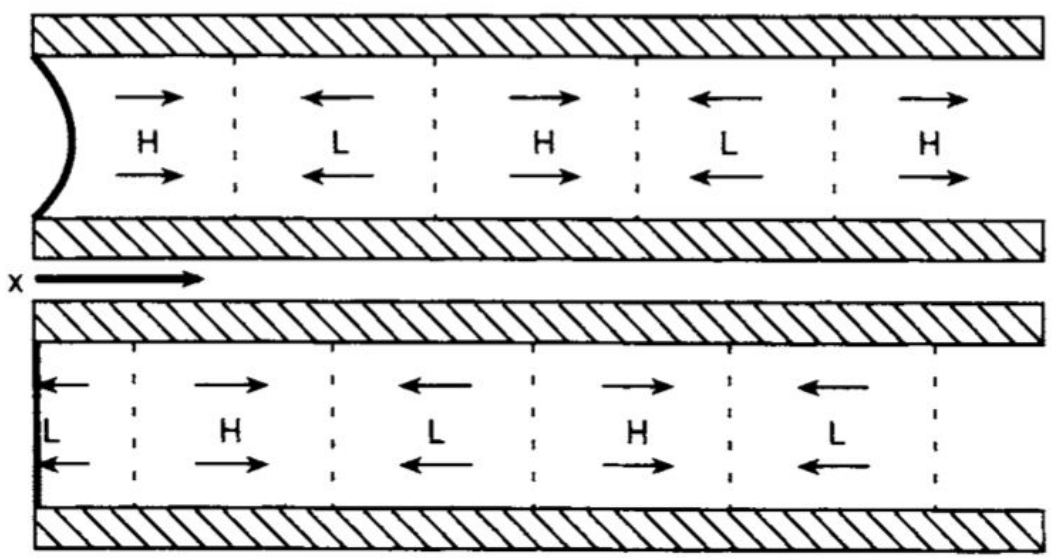
Acoustic (sound) waves

Acoustic waves do not play an important role in setting the physical properties of the ocean, but have uses in a number of related applied settings (sonar, seismic imaging,...). In the atmosphere, they can play a more important role, and are able to drive pressure and temperature fluctuations in the middle/upper atmosphere.

Acoustic waves are able to propagate because of the compressibility of the medium in which they exist. The atmosphere is primarily composed of gas, and so is compressible. In contrast, the ocean is liquid, and so is practically incompressible. Nevertheless, sound waves do also exist in the ocean.

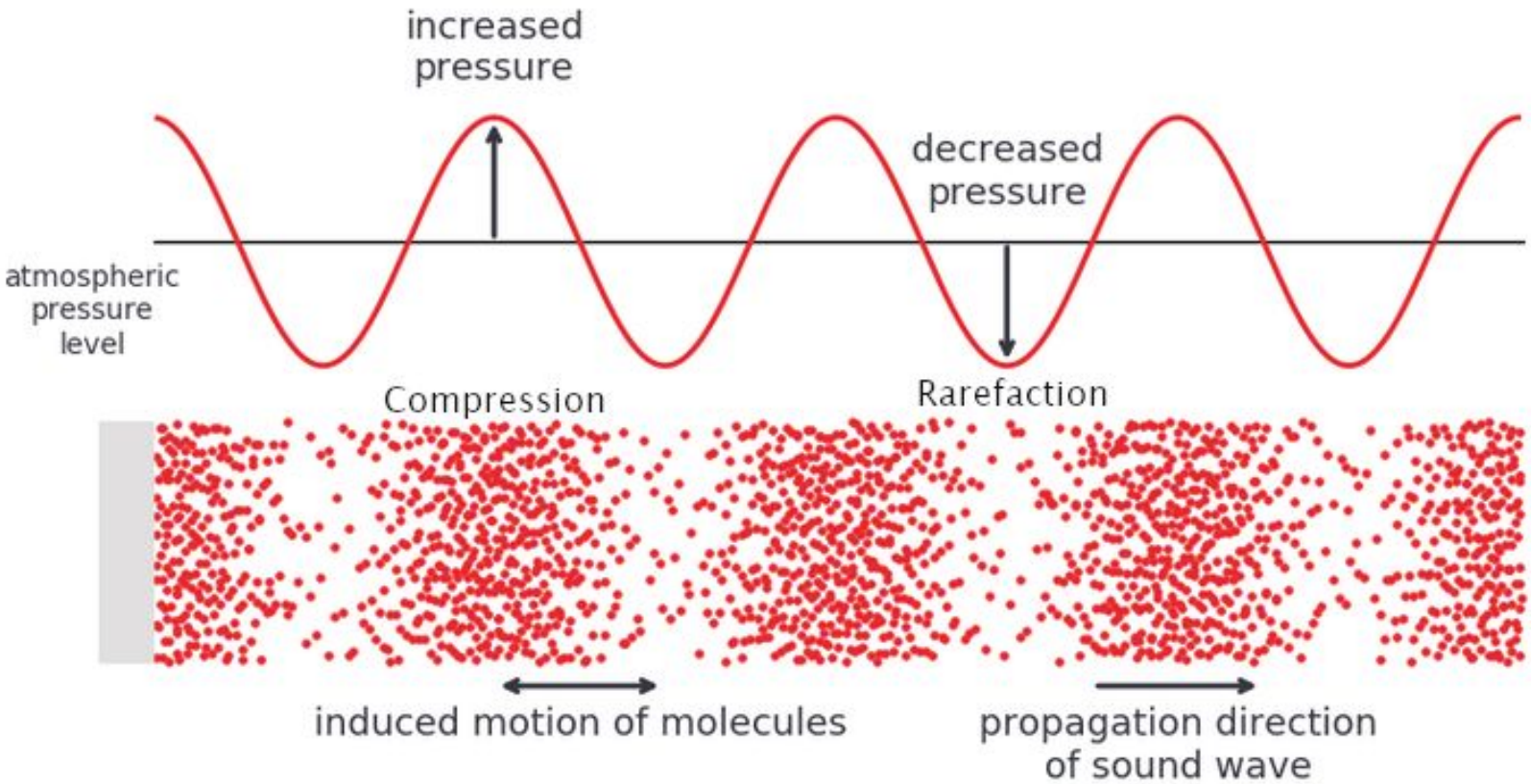


Acoustic waves are different from the other waves that we have seen until now in that they are purely longitudinal: the oscillations are parallel to the direction of propagation.

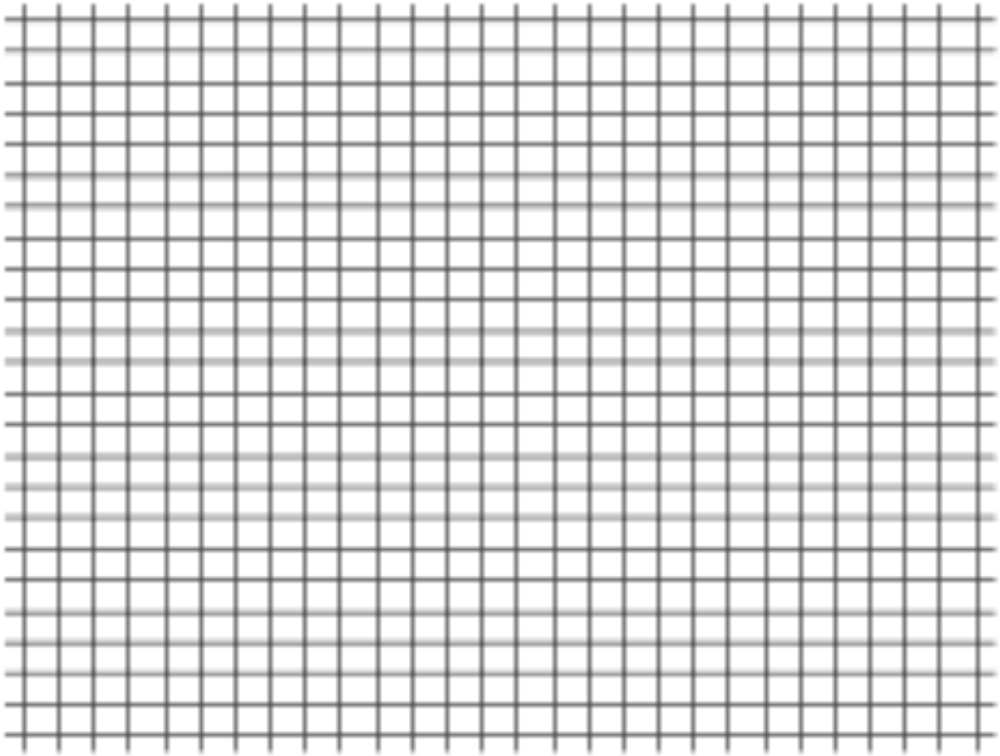
Properties of acoustic waves

Acoustic waves are compressional waves: they exist because water is (slightly!) compressible. Physically, they represent rapid density oscillations due to pressure changes in an elastic medium.

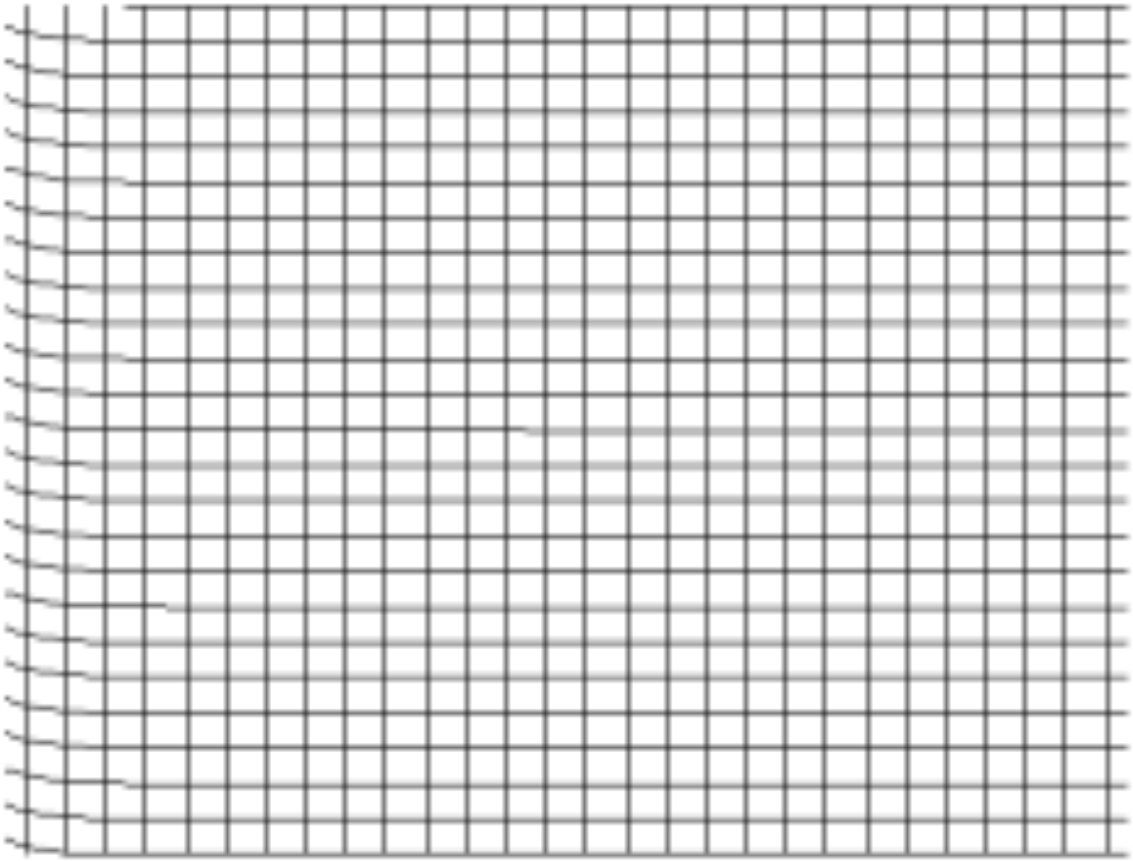
They are purely longitudinal waves: the displacements of the particles as the wave propagates are parallel to the direction of propagation.



Longitudinal waves:



Transversal waves:



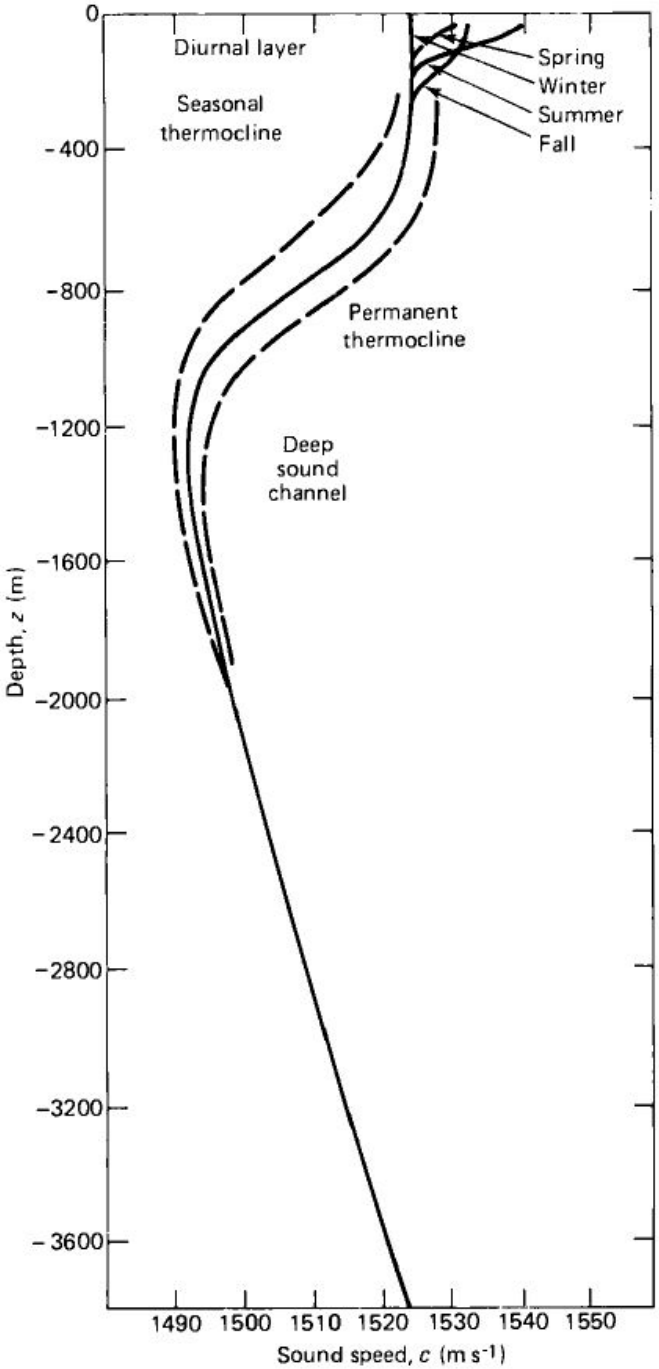
Water properties' influence on the speed of sound

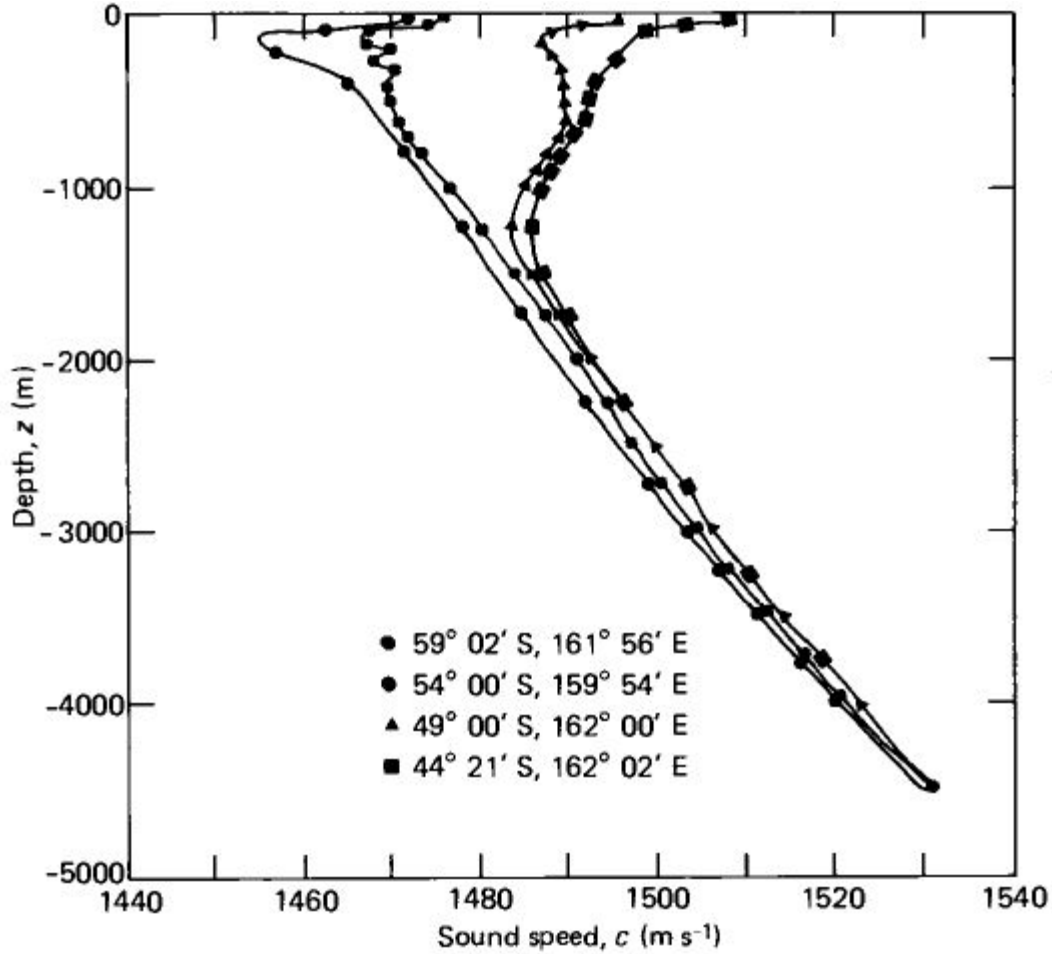
$$c = c(S,T,z) = c_0 + \alpha_0(T-10) + \beta_0(T-10)^2 + \gamma_0(T-18)^2 + \delta_0(S-35) + \epsilon_0(T-18)(S-35) + \zeta_0|z|$$

The relative influence of the various terms depends on the location of the wave within the water column:

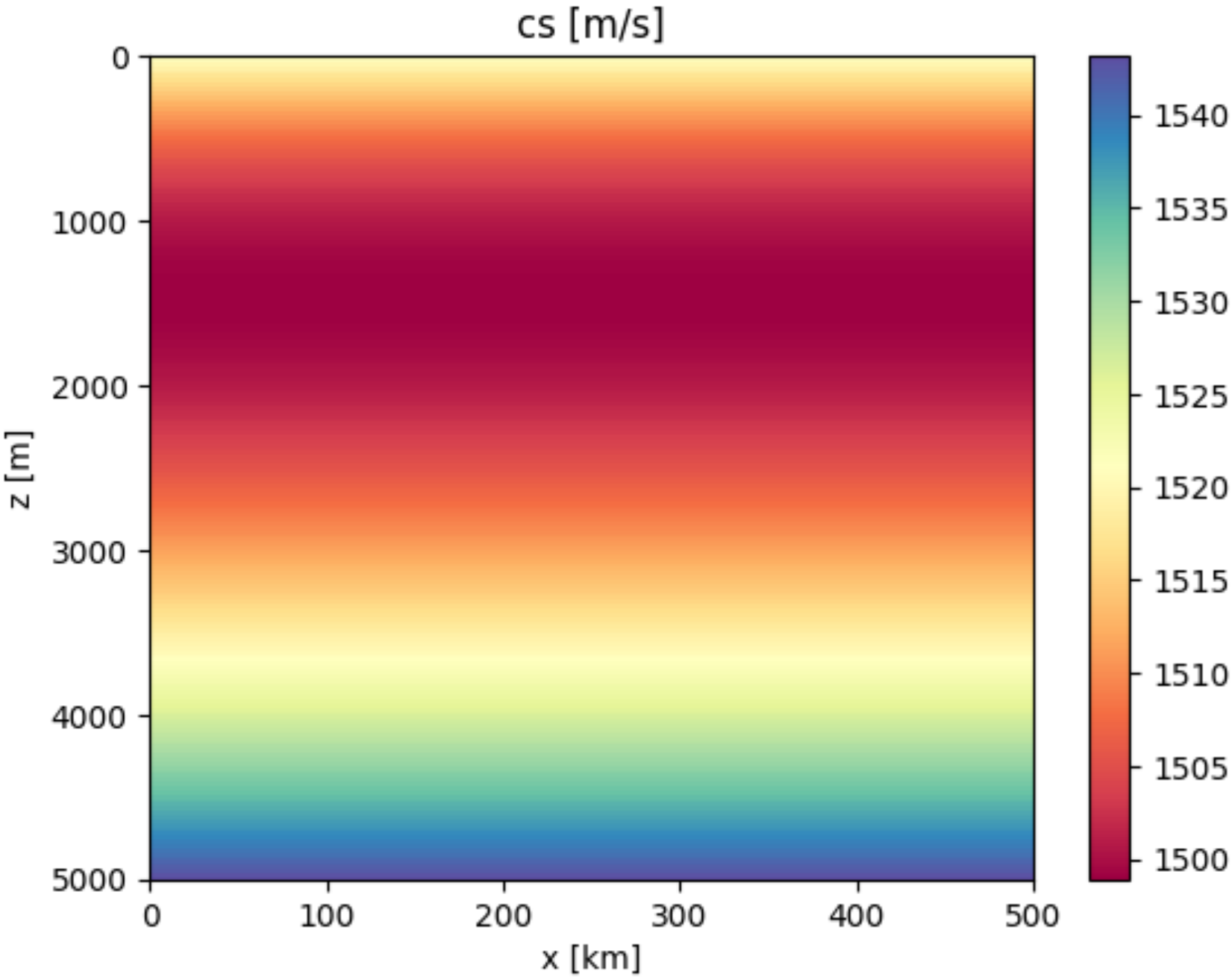
- A temperature rise of 1°C causes an increase in speed of ~3 m/s
- An increase in salinity of 1 causes an increase in speed of ~1.1 m/s
- In deep water, the effects of depth dominate: sound speed increases at a rate of about 1.64m/s for every 100m increase of depth
- In shallow water, the effects of temperature dominate: the characteristics of the near-surface mixed layer and the thermocline are very important in establishing the propagation of acoustic energy in the upper levels of the ocean

This dependence on temperature, salinity and depth means that wave speed can also change as the wave moves from one part of the ocean to another. This can lead to refraction of the wave, in the same way as for a surface gravity wave entering shallower waters.





Sound speed profiles taken near Bermuda (left) and in the Pacific (above), along the 160°E longitude line. The minima in the profiles lead to the development of sound “channels” that can trap acoustical energy and allow it to propagate over long distances.
This channel shoals in the subpolar regions, or can disappear.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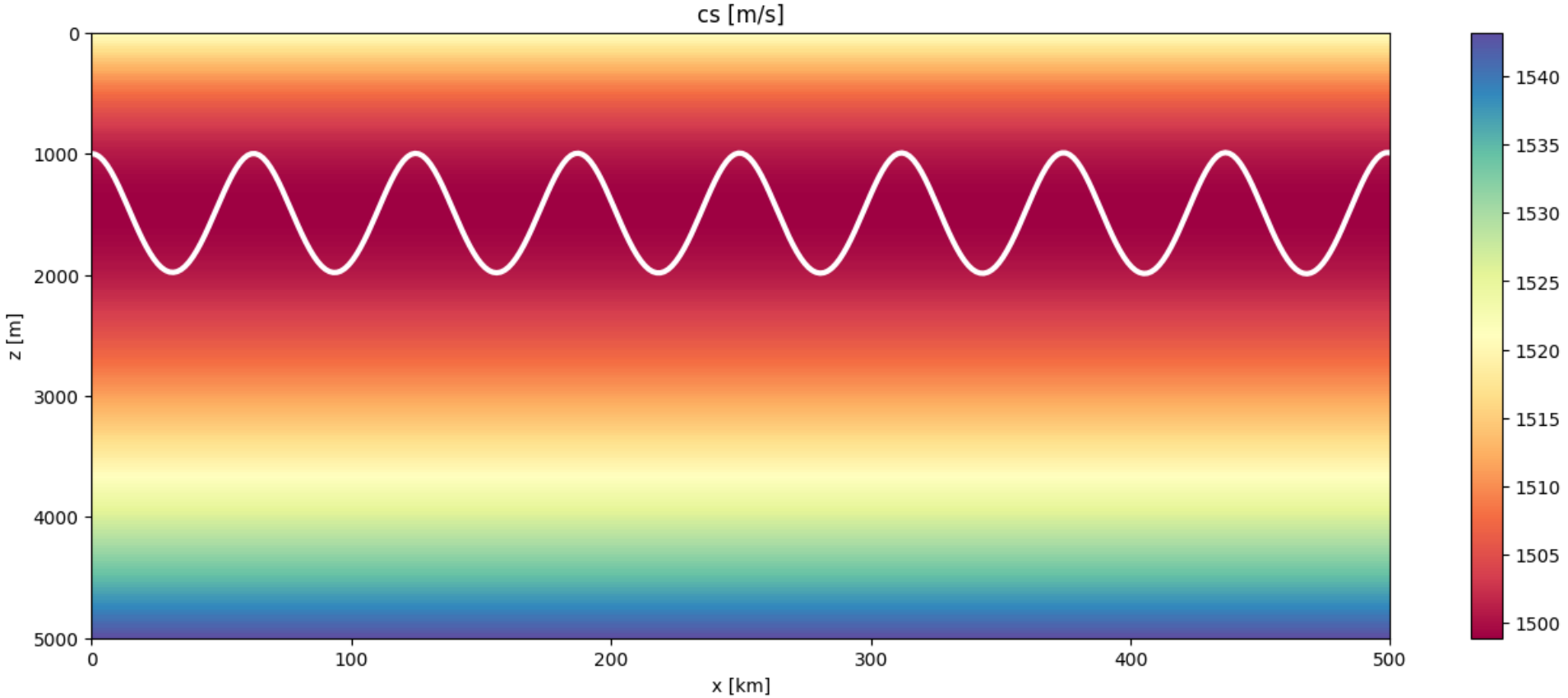
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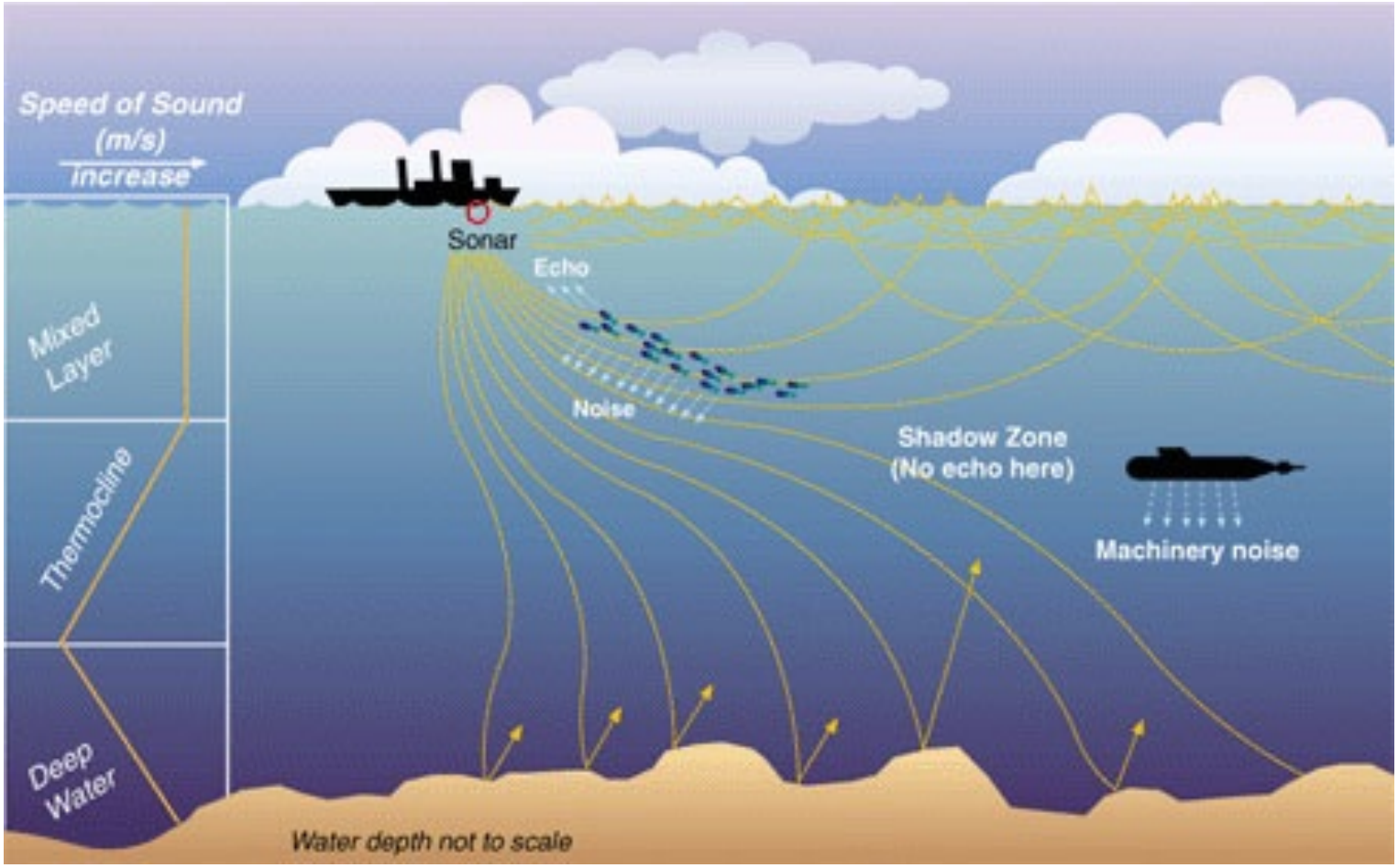
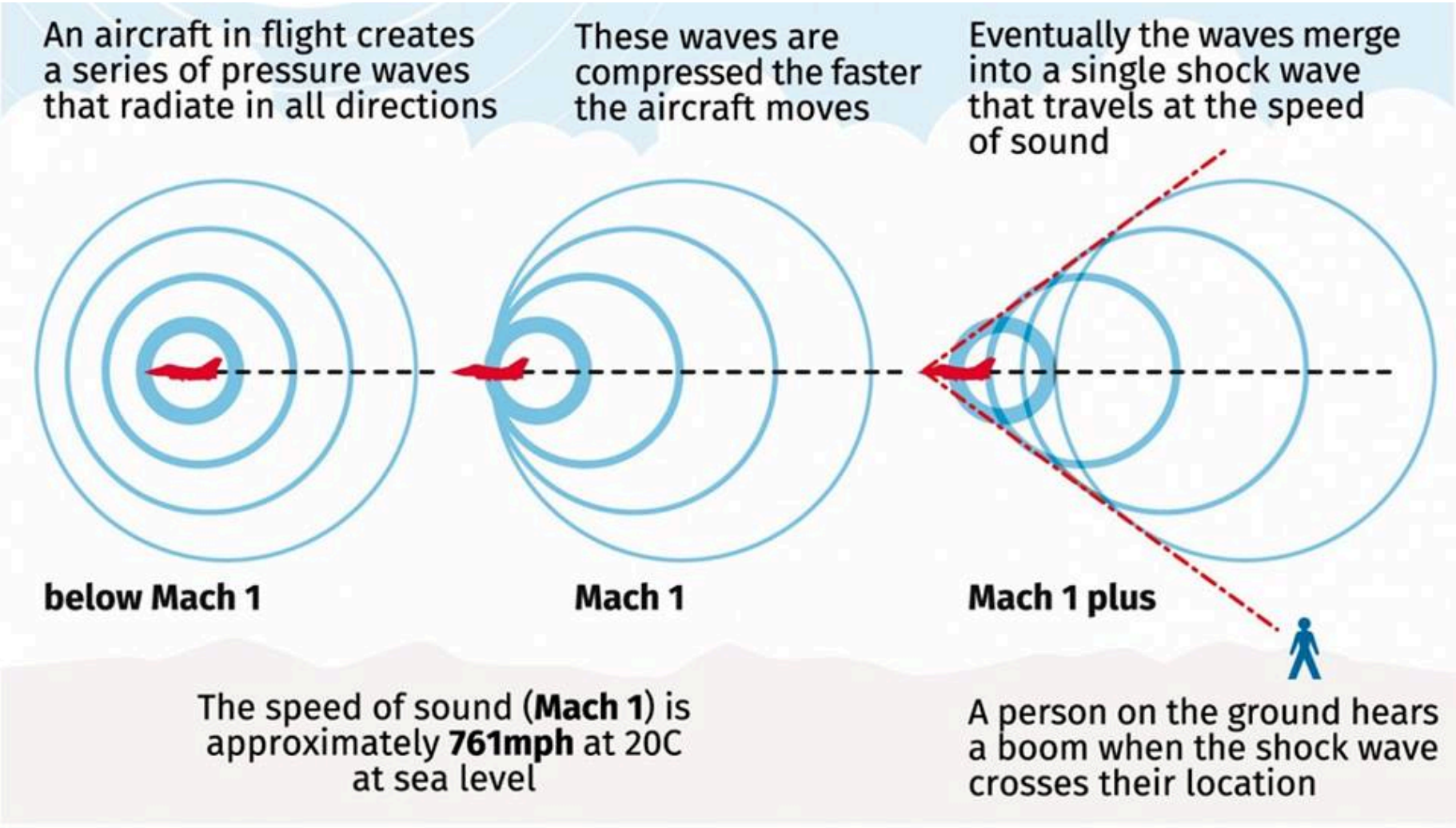
Acoustic Waves



Sonic Boom

Sonic Boom


An aircraft in flight creates a series of pressure waves that radiate in all directions
These waves are compressed the faster the aircraft moves
Eventually the waves merge into a single shock wave that travels at the speed of sound
below Mach 1
Mach 1
Mach 1 plus
The speed of sound (Mach 1) is approximately 761mph at 20C at sea level
A person on the ground hears a boom when the shock wave crosses their location

Sonic Boom